Low-Cost Helper-Carrier Reception Across Random Frequency Hops

Methods, Results, and Recommended Practices

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Abstract. A narrowband reference can provide precise frequency, phase and sampling-clock estimates for subsequent weak-signal reception. This article presents the methods, experimental results and recommended practices developed for the standalone Helper Carrier receiver. The principal condition is a 40 W Helper signal with 180 dB path loss, cryptographically randomized 30 MHz frequency hopping, a 36 s estimation interval and a fixed 38 s lead-in. Frequency-aided correlations, covariance-aware clock fitting and dual-pilot phase combination provided accurate, inexpensive clean-hop tracking; joint-dwell estimation produced a further HF improvement. In the latest 32-case evaluation, frozen frequency estimates were available in 31 cases, with 0.01357 Hz RMS error, 0.02887 Hz worst error and a median conditional half-width of 0.11695 Hz; all 31 reported intervals covered. The approximate ±0.1 Hz development objective was an aspirational guide to useful search-space narrowing, rather than a pass/fail requirement. Complete-dwell conditioning corrected and released every captured interval while preserving capture timestamps. Pooling auxiliary phase-inversion evidence across independently phased dwells achieved 0.441 ms timing RMS and 0.965 ms worst timing error, reduced wrong-hop assignments by 54.1%, and reduced complete-output clean-twin phase RMS from 8.843° to 6.859°, without reducing output. Tracking consumed 0.336-0.374 CPU s per 36 s; complete 40 s conditioning and drain used a median 0.703 CPU s. An archived cable experiment provided additional evidence of effective correction and was closely reproduced by a compact simulation. The results establish useful standalone synchronization and conditioning techniques. Comparisons with alternative implementations identify their tradeoffs and operating limits, supporting practical design recommendations and a concrete basis for downstream integration.

Keywords: *Carrier synchronization; frequency-hopping spread spectrum; sampling-clock estimation; weak-signal timing; delayed baseband conditioning; uncertainty; software-defined radio.*

1. Introduction

Helper Carrier is an experimental three-tone reference for making weak-signal acquisition less expensive and less ambiguous. It estimates clock and carrier state and supplies corrected baseband. Its eventual beneficiary is the ordinary Robust receiver; the work evaluated here establishes the standalone assistance techniques without requiring auxiliary payload or Robust decoding [4-6].

Random frequency hops introduce fresh RF phases, while independent audio and RF oscillators contribute different errors. Retuning may disturb only part of a dwell, and boundary errors can assign an accurate phase estimate to the wrong samples. Effective assistance therefore requires both precise estimation and correct application across the captured stream.

This paper examines which estimators improve phase and frequency measurements, which methods improve complete output and preserve timing, and which practices balance precision, availability and computational cost. Successful techniques form the basis for recommended practices; comparisons and operating limits explain the design choices. Separate campaigns retain their individual experimental scope.

The approximate ±0.1 Hz target was an aspirational development guideline, not a mandatory acceptance criterion. Useful search-space narrowing was achieved, including where a conservative reported half-width slightly exceeded 0.1 Hz. Downstream decoding and its exact search budget remain integration questions.

Finite-record sinusoidal estimation, leakage-efficient weighting and variable-delay interpolation provide the signal-processing background [1-3]. The contribution is an experimental evaluation of their practical use in a weak-signal receiver with independently phased hops.

2. Experimental methods and signal model

2.1. Reference waveform and observability

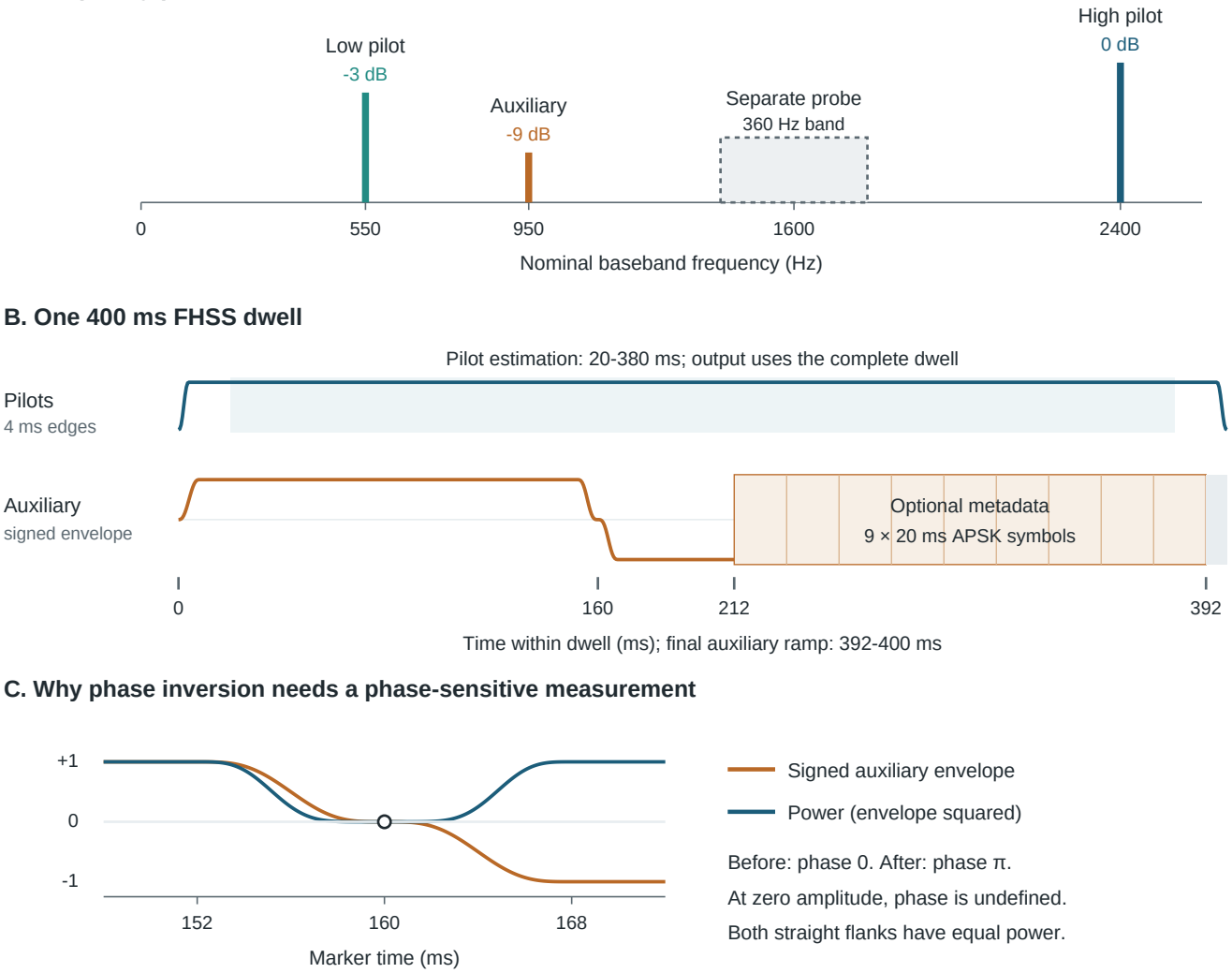


Figure 1. Helper format in the studied 2400 Hz high-pilot configuration. A: nominal frequency positions and steady-tone power ratios; stems are schematic, not a measured spectrum. B: one FHSS dwell, its guarded pilot-estimation aperture, auxiliary reversal and optional metadata interval. The drawing does not make the aperture an output mask. C: signed auxiliary envelope and power through the reversal, using the specified septic taper. The phase changes by π through a zero-amplitude point; equal power before and after the event does not reveal the sign change in a magnitude-only waterfall. Metadata-dependent complex values are not plotted as a fictitious straight tone.

The nominal baseband waveform contains pilots at 550 and 2400 Hz and an auxiliary component at 950 Hz. Relative to the high pilot, the low pilot is 3 dB lower in power and the auxiliary is 9 dB lower before the waveform normalization. Helper power refers to the total normalized Helper waveform, not to each tone. A dwell lasts 400 ms. The auxiliary undergoes a known reversal centered at 160 ms, tapering through zero over 152-168 ms; the two pilots do not reverse. The waveform and auxiliary payload format were unchanged by the pooled-timing experiment [6,12].

Figure 1 gives a visual reading of the format. The two straight pilots provide frequency and relative-clock evidence. The auxiliary has a different role: its known reversal supplies timing evidence, while its later symbols carry optional metadata. In a complete metadata-bearing dwell, nine 20 ms 16384-APSK symbols occupy 212-392 ms. They are auxiliary metadata, not the separate 360 Hz shaped diagnostic 64-APSK probe used in the experiments. Frequency assistance and reversal timing do not require decoding those symbols. In the current waveform, FHSS pilot boundaries are shaped over 4 ms on either side, whereas the auxiliary has 8 ms boundary ramps; with FHSS disabled, the straight pilots remain continuous across dwell boundaries [12].

An affine frequency model separates the received sampling-clock scale, a, from the common frequency translation, b. With received pilot frequencies ν_L and ν_H, the nominal relationship is

$$\nu_L = a\, f_L + b,\ \nu_H = a\, f_H + b. \quad (1)$$

$$a = (\nu_H - \nu_L)/1850,\ b = \nu_H - 2400a. \quad (2)$$

The difference of the two measured phases cancels a common RF hop phase and supplies precise relative-clock information. It retains an unknown constant differential phase and an integer-cycle ambiguity. At 1850 Hz pilot separation, one differential phase cycle corresponds to approximately 540.5 µs. An arbitrary change in differential channel phase can imitate a change in clock timing. Neither repeated observations nor buffering removes this identifiability problem without additional evidence.

Every new hop is therefore assigned a fresh RF phase. Historical information constrains frequency and clock evolution, but RF phase is not assumed continuous across hops. The receiver preserves one initial relative timing/phase reference through subsequent observations and gaps. Absolute source-time ambiguity is reported rather than silently removed by a new reference.

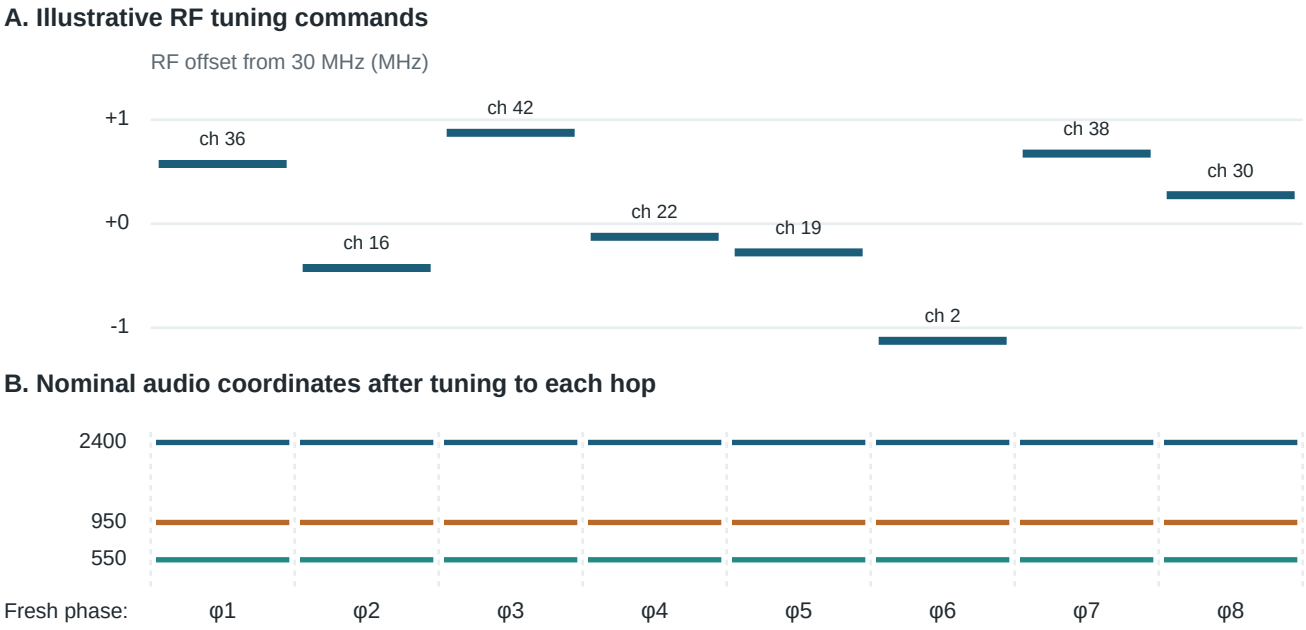


Figure 2. RF hopping and audio-baseband coordinates are different views of the same signal. Top: an illustrative cryptographic-stream channel sequence on the 50-channel, 50 kHz HF grid; each bar represents the narrow Helper group, whose internal tone spacing is unresolved on this RF scale. Bottom: after tuning to each commanded hop, the nominal pilot locations remain near 550, 950 and 2400 Hz, but the common phase can change independently on every dwell. The example is explanatory, not a captured waterfall or a performance trial.

Table 1. Principal complete-stream experiment. These are simulation conditions, not a certified radio installation.

Quantity	Configuration
Helper / probe power	40 W Helper; separate 40 W probe during 38-40 s. Simultaneous transmission is 80 W combined.
Propagation and receiver noise	180 dB principal path loss; screens at 175, 170, 165 and 160 dB; 290 K thermal noise; 10 dB noise figure.
FHSS and sampling	50 channels at 50 kHz spacing around 30 MHz; cryptographic-keystream schedules; 400 ms dwell; 24 ksample/s complex capture.
Oscillators	Independent +37/-13 ppm audio-clock settings; 0.8 ppm relative RF error; modeled drift and phase noise.
Observation chronology	Freeze at exactly 36 s; fixed 38 s Helper lead; held-out tone or 360 Hz shaped APSK capture through 40 s; drain buffered output afterward.
Pilot aperture and delay	Settled pilot aperture 20-380 ms into a nominal dwell; default complete-dwell delay 2 s; comparisons at 0.4, 1 and 3 s.
Matched cases	Same impaired capture for each pair; eight schedules per loss screen; 32 fresh reserved 180 dB holdouts for the pooled-inversion finalist.

2.2. Sampled link and oscillator simulation

The principal simulation generates the actual Helper transmitter waveform in complex baseband, using 9600 nominal samples per 400 ms dwell at 24 ksample/s. Separate transmitter and receiver audio clocks determine source time and physical ADC time. A relative RF error is multiplied by the current commanded RF frequency and integrated over physical time; changing the RF command therefore changes its frequency contribution. At each random hop an independent random phase increment is added. Brownian RF phase noise has a nominal scale of 0.05°/√s. Clean fixtures set clock and RF drift to zero; separate controls introduce drift, steps, curvature, retune displacement and other departures [8,10,11].

The Helper amplitude is normalized from its total active-mean waveform power, including its actual pilot/auxiliary allocation. A separately specified probe is mixed into the same channel. Power attenuation L dB gives $P_{rx} = P_{tx}10^{-L/10}$. With input temperature T and noise figure NF, the noise density is $N_0 = k_B T 10^{NF/10}$. In the implementation the straight high pilot is normalized to unit complex amplitude. For its linear C/N_0 value γ_H, the independent Gaussian I/Q variances are

$$\gamma_H = P_{H,rx}/N_0,\ \sigma_I^2 = \sigma_Q^2 = F_{ADC}/(2\gamma_H). \quad (3)$$

F_{ADC} is the physical sample rate, including the modeled receiver clock error. This convention preserves total complex noise power $F_{ADC}N_0$ before normalization. The reference conditions are 290 K and 10 dB noise figure. Thermal-noise seeds are separate from the schedule/oscillator stream, so changing noise need not change the random hops or phase-noise history. This is a flat-channel, narrowband simulation with selected impairments, not a complete fading, multipath or lunar-scattering model.

Impairment fixtures distinguish actual common signal phase/frequency changes from contamination applied only to a pilot. Missing ADC samples are separate from an RF outage during which the ADC still captures noise. Retune dates can be displaced independently from the transmitted dwell calendar. Comparators consume identical impaired samples within a trial; wrapper comparisons across different physical models are not automatically identical-noise comparisons. Exactly 90 dwell intervals determine the frozen 36 s result, the transmit lead lasts 95 dwells, and capture ends after 100 dwells. Buffered output may drain later.

2.3. Separation of receiver evidence and evaluation truth

The tone and 360 Hz shaped APSK probes are passive diagnostics. They provide no synchronization, correction coefficients, candidate selection, thresholds or adaptive alignment. Evaluation retains one constant reference; it does not fit a new phase, gain or timing reference at each hop. Clean-twin results isolate correction error from probe noise. Noisy-probe results include that noise and residual components. Oracle diagnostics additionally know nuisance quantities unavailable to the receiver and are reported separately [6-8].

Complete 0-40 s phase metrics use an evaluator-only continuous synthetic clean twin passed through the measured correction path. Before 38 s, this diagnostic does not add a transmitted probe to the receiver capture. The independent powered tone or APSK probe occupies 38-40 s. Thus complete-stream correction diagnostics and held-out noisy-probe measurements have different signal scopes, even when they use the same receiver history.

Figure 3 provides an observational counterpart to the format diagram: a short-time Fourier transform of the archived analog cable recording, computed for this article without making a new capture. Its horizontal coordinate is received audio frequency and its vertical coordinate is original capture-file time. The low pilot, auxiliary and high pilot form distinct tracks; the diagnostic tone appears near 1604 Hz only near the end. This record is a strong, non-FHSS audio-path example with deliberately imposed clock and frequency errors. It is not the 180 dB RF simulation or evidence that a per-dwell marker is visible at that loss. The zoom shows temporal structure at reduced frequency resolution; neither display measures the inversion sign directly.

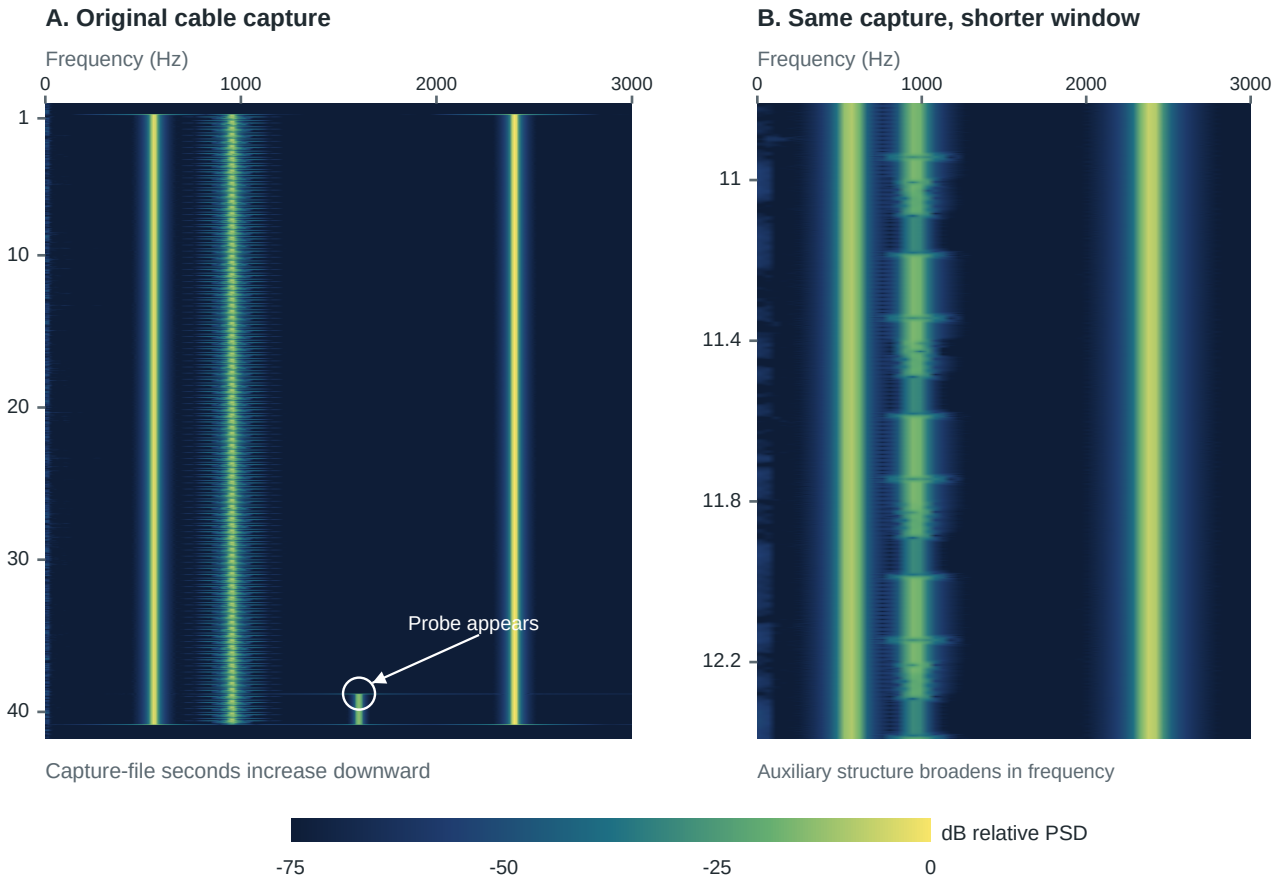


Figure 3. Waterfalls from the archived 48 ksample/s mono cable recording [10]. A: the complete 41.7773 s capture, containing a nominal 40 s stimulus plus capture preroll/postroll; 4096-sample Hann windows (85.33 ms), 20 ms hop, 11.71875 Hz FFT spacing. B: 10.8-12.4 s of the same capture; 1024-sample Hann windows (21.33 ms), 2 ms hop, 46.875 Hz spacing. Both panels use the same PSD reference and -75 to 0 dB color range. Time increases downward; data are uncorrected and not aligned using the probe. The finer time window broadens the spectral tracks. This historical recording does not qualify the current optional no-ramp waveform.

The frequency result labeled available at 36 s is frozen at that capture frontier. Fresh phase measurements on subsequent hops, and any later clock or calendar information used to correct buffered samples, are separate results. Each released block carries both its original capture dates and the latest Helper observation used. This distinction prevents retrospective correction from being presented as causal information available earlier.

Screening and holdout schedules used distinct verified 32-byte keys and distinct noise seeds. This matters because different numerical seed labels in earlier fixture generators could repeat a key. Freshness here is relative to the recorded reservation inventory, not to every unlogged experiment. Once examined, the published holdouts are regression cases. Recorded impairment tests include retune errors, pulses, additive contamination, missing input, frequency steps and curvature, differential-phase violations, clock-cycle jumps and noise-only controls [6,7].

2.4. Diagnostic tone, APSK and complete-stream scoring

The diagnostic alternatives are an unmodulated 1600 Hz tone and 64-APSK centered at 1600 Hz, sent at 300 symbols/s with root-raised-cosine rolloff 0.2. The nominal shaped bandwidth is therefore 300(1 + 0.2) = 360 Hz. The 64-point constellation has four rings containing 4, 12, 20 and 28 points, with radii proportional to 1, 2.8, 4.6 and 6.4 and normalization to unit average symbol energy. The simulator interpolates the pulse on a 1/4096-symbol grid and truncates it to eight symbols on each side. Finite pulse endpoints and interpolation dependencies affect which symbols can be evaluated [8,10].

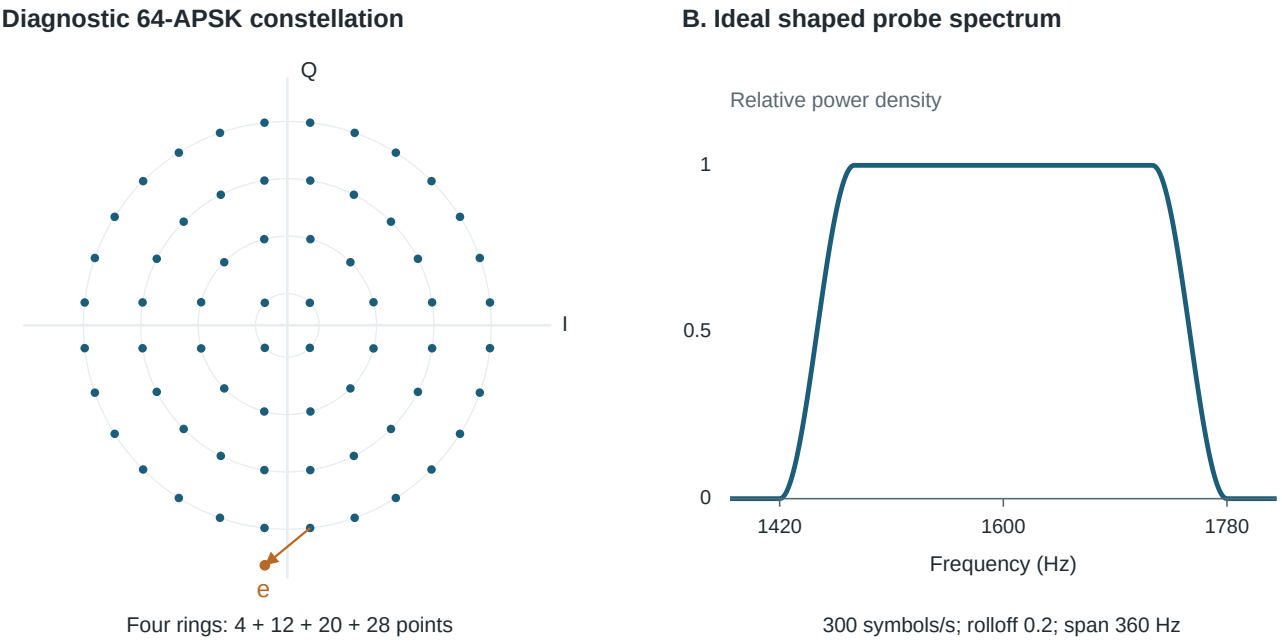
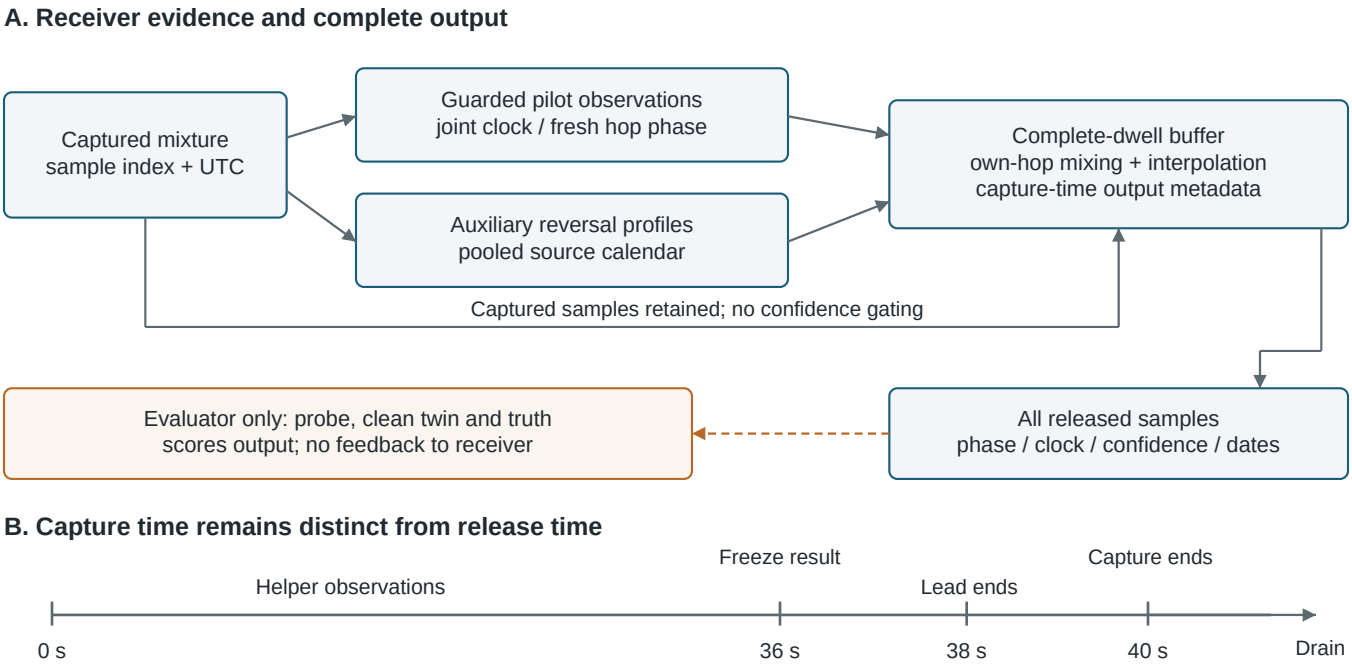


Figure 4. Diagnostic 64-APSK probe, distinct from the auxiliary metadata. A: the unit-average-energy constellation used by the fixture; the example error vector illustrates EVM geometry and is not a measured symbol. B: ideal transmit power-spectrum shape for 300 symbols/s and root-raised-cosine rolloff 0.2, centered at 1600 Hz; the ideal occupied span is 1420-1780 Hz. The finite pulse implementation approximates this shape. The evaluator uses fixed source dates and one retained reference, so constellation alignment cannot conceal a receiver timing or phase error.

The APSK evaluator applies a fixed matched filter at known source symbol dates and amplitude after the Helper-derived correction. It does not search probe timing, frequency or phase, select a clock branch, or optimize gain. With expected symbols s_k and errors e_k at those fixed dates, its metric is

$$\mathrm{EVM}_{\mathrm{rms}} = \sqrt{(\Sigma|e_k|^2/\Sigma|s_k|^2)},\ \mathrm{S/E} = -20\log_{10}(\mathrm{EVM}_{\mathrm{rms}})\ \mathrm{dB}. \quad (4)$$

S/E is a constellation signal-to-error ratio, not automatically thermal SNR: timing, phase, amplitude distortion and residual Helper components also contribute. Noisy and clean-twin EVM use the same receiver correction. The tone is scored by fixed-window residual phase/frequency fits, with conditional fit errors; unavailable noisy fits are reported as such. Complete-stream phase metrics include uncertain output and separately identify undefined phase at zero clean energy. Sample regions, common-support intersections, excluded symbol footprints and physically missing samples are reported rather than silently removed from denominators.



Timeline is schematic; later evidence may correct only output still buffered.

Figure 5. Standalone receiver and evaluation separation. Pilot and auxiliary observations control correction; the probe and simulator truth only score the released output. A fresh phase is estimated for each hop. The 36 s snapshot is independent of later delayed correction. The diagram is schematic, not a claim of downstream integration.

3. Receiver strategies

3.1. Frequency-aided pilot estimation and joint clock history

Broad acquisition is retained for startup and inconsistent predictions. Once acquired, the accumulated frequency and clock estimate places inexpensive local correlations near the expected pilot frequencies. The baseline uses guarded Hann-weighted observations and a four-sub-aperture frequency slope. Later implementations include covariance-aware local summaries, a joint fit to frequency evidence and pilot phase-difference evolution, and combination of both phases for the current common RF phase [4-6].

A later actual joint-dwell path uses current-dwell complex evidence together with prior relative-clock information to estimate the fresh phase and residual frequency. Its prediction is evaluated before the current dwell is ingested. Shared observations and conditioning are not counted as independent new measurements. This is a stronger implementation test than an offline fit that profiles independent pilot gains and combines their phases afterward without updating receiver history. Nevertheless, the measured benefit is conditional on the tested oscillator and differential-channel models; mismatch requires fallback or reduced authority.

3.2. Complete-dwell conditioning and capture-time preservation

The complete conditioner holds captured dwells until their own Helper observations and available later history have been processed. It extrapolates the applicable frequency, phase and clock model before and after the guarded pilot aperture. The critical implementation rule is to apply each input sample's own-hop phase before interpolation, rather than interpolate a random phase discontinuity and then apply one correction. The current implementation uses two-point interpolation with explicit dependency handling [6,8].

All captured samples are processed and the corrected/resampled output is released chronologically. A conservative well-supported flag is diagnostic; it is not an instruction to blank, fade or omit uncertain output. Actual missing input is represented as a discontinuity and is not silently concatenated or replaced. Tests inspect the complex values before, inside and after the aperture, including extrapolation sign and date, rather than checking only an enlarged support mask.

Output metadata distinguish fractional input sample coordinate, capture UTC and monotonic time, clock epoch, release time, relative source-time mapping and latest Helper evidence. A standalone consumer performs time-dependent pattern lookup from capture time, not release wall time. Startup, drain, irregular chunks, missing input and clock discontinuities are exercised. These tests establish the software mapping in the standalone model; they do not establish hardware timestamp accuracy or interoperability with Robust.

3.3. Pooling weak auxiliary-inversion evidence

The strict auxiliary detector was accurate when it accepted a strong individual marker, but usually rejected weak dwells. The more sensitive experimental path retains an entire weak timing likelihood instead of requiring each dwell to cross that detector's threshold. It mixes an 80 ms window around the known reversal using pilot-aided frequency coordinates, projects out the neighboring pilots, and forms disjoint 250 µs complex correlation bins. A 129-point grid at 125 µs spacing covers the existing ±8 ms calendar prior. Three residual-frequency hypotheses check prediction mismatch [6,8].

For a trial timing δ and residual frequency ρ, let y be the correlation vector, s the reversal template and P the projection that removes the modeled neighboring components. Profiling one unknown complex auxiliary gain gives a score of the form

$$Q(\delta, \rho) = |s^{H}Py|^{2} / (s^{H}Ps). \quad (5)$$

The implementation compares reversal evidence with a straight auxiliary tone and estimates noise from received residuals. A different complex gain is fitted in every dwell, so random hop phases are never coherently joined. Up to 96 disjoint profiles are mapped under the retained two-pilot clock and pooled once. Repeated queries do not create more observations; shared clock and model allowances are not divided by the number of dwells. Early/late groups can reject contradictory maps when each has sufficient evidence.

The resulting observation is a repeated source-calendar estimate modulo 400 ms. It is not an absolute UTC decoder and does not measure arbitrary receiver RF retune delay. The complete-stream adapter applies qualified calendar adjustments within the original boundary prior while preserving the physical-boundary uncertainty enclosure. It introduces no new clock feedback loop or reference reset. Its value must therefore be measured by actual sample-to-hop association and complete output, not by timing precision alone.

4. Results: techniques that produced useful gains

4.1. Joint history and dual-pilot phase combination

An earlier eight-case comparison explains why frequency-aided tracking became the foundation of subsequent work. Broad-only acquisition, aided phase alone, and aided phase plus relative-clock tracking had median clean-twin phase RMS of 6.54°, 10.55° and 3.18°, respectively. Aiding the current phase measurement did not by itself improve fixed-reference output; correctly tracking relative time was essential. On the same captures, broad-only processing required 0.392-0.437 CPU s through 36 s, versus 0.060-0.069 s for the aided-plus-clock path. These are matched historical measurements, not a comparison with the more elaborate current pipeline [4].

The early matched 32-case ablation in Table 2 separates measurement accuracy from interval width. Incorporating covariance and the phase-difference clock history into the joint solution substantially narrowed the reported frequency interval while leaving point-error RMS approximately unchanged. Combining both pilot phases improved phase error. The partitioned-Hann alternative did not establish a material further gain [6].

Table 2. Historical matched HF comparison at 40 W / 180 dB, 32 cases. Phase entries are median/worst per-case clean-tone RMS, not pooled sample percentiles. All 32 frequency intervals covered for each method.

Method	Frequency RMS (Hz)	Median half-width (Hz)	Phase RMS (°)	Full windows
Four-window baseline, relative clock	0.01658	±0.21393	3.08 / 6.34	160/160
Four-window covariance + joint fit + dual phase	0.01762	±0.12211	2.30 / 4.44	159/160
Partitioned Hann + joint fit + dual phase	0.01783	±0.12520	2.29 / 4.80	158/160

Covariance alone changed neither the point estimates nor interval widths in that comparison; the later joint fit made the difference. For the four-window joint/dual path, median modeled thermal frequency sigma was 0.02445 Hz, common-process sigma approximately 0.000059 Hz, bounded-model allowance 0.00121 Hz, and the recent-half enclosure contributed 0.04553 Hz. These are accounting components of a conditional engineering interval, not independent demonstrations of calibration. The observed approximately 0.018 Hz error is therefore not evidence that the larger enclosure can simply be removed.

The phase benefit also survived a more direct ablation: within the partitioned joint path, high-pilot-only and dual-pilot phase estimates had median RMS 3.18° and 2.29° at the same 158/160 supported windows. In a separate 32-case APSK comparison, joint high-pilot-only versus dual phase changed median clean-twin EVM from 5.585% to 4.183%, whereas noisy EVM changed only from 55.158% to 54.928%. The limited noisy improvement is consistent with a probe dominated by received noise; it is not evidence that the cleaner phase estimate has no value [6].

4.2. Actual joint-dwell estimation: a modest HF improvement

The v5 joint-dwell candidate updated the actual receiver chain and retained the baseline for paired evaluation. On 32 fresh HF cases, pooled common-dwell phase RMS improved from 2.886° to 2.623°, and median common-supported clean-sample phase RMS improved from 2.447° to 2.073°. Fully supported later windows increased from 140/160 to 154/160; windows with any support increased from 157/160 to 159/160. Thus its gain was not produced solely by rejecting harder samples. Worst common-dwell phase fell from 11.093° to 9.071° [6].

The same comparison changed 36 s frequency-error RMS only from 0.01302 to 0.01279 Hz. Median conditional half-width was 0.11821 Hz for baseline and 0.11670 Hz for joint-dwell. Both supplied useful, approximately tenth-hertz narrowing while retaining allowances beyond the much smaller measured point errors. Mean receiver process CPU through 36 s increased from 0.202 to 0.321 s. In eight HF APSK cases, common-symbol clean EVM changed from 4.59% to 4.04%, while noisy EVM changed from 54.59% to 54.50%. These are useful but modest improvements.

Transfer to the earlier 5.8 GHz fixtures was not uniformly favorable. Supported windows decreased from 38 to 26 in the FCC100 fixture and from 40 to 28 in EU20. A current-dwell fit that helps one operating geometry is therefore not established as the best estimator for all geometries. The selected HF chain uses joint-dwell estimation, but the evidence does not justify removing selectable alternatives.

4.3. Pilot-envelope boundary tracking: useful but conditional

The unsuccessful five-boundary ambiguity experiment discussed in Section 5 must be distinguished from the later v9 ramp-based boundary tracker. The latter ran in the actual complete-stream chain. It scored 65 candidate offsets within a supplied ±8 ms prior and retained at most 16 profiles. Independent settled pilot gains were correlated on each side of a boundary, without imposing RF phase continuity. Adjacent profiles share noisy templates, so their curvature was not treated as a calibrated timing interval [6].

On 32 fresh matched HF cases at 40 W / 180 dB, nominal versus ramp-tracked complete-output clean-twin phase RMS was 8.66° versus 4.76°. Beginning/interior/end RMS changed from 20.25°/3.90°/4.37° to 7.98°/3.90°/4.02°; the combined boundary region changed from 14.65° to 6.32°. Wrong physical-hop assignments fell from 167,170 to 20,725 while both methods emitted 30,721,548 resampled outputs. All captured input remained represented. Thus this was a practical boundary-assignment gain, not simply a wider support mask or a claim that clock-cycle ambiguity had been solved.

The limitations were substantial. Worst output phase remained approximately 180°. A physical retune 2 ms early worsened held-out boundary RMS from 7.74° to 12.13° relative to nominal scheduling. In eight APSK cases, mean clean-twin EVM improved from 6.93% to 5.09%, while noisy EVM changed from 55.12% to 54.93%. These results motivate retaining the method as an experimental alternative when ramp structure exists; they do not validate it without ramps or establish a calibrated boundary enclosure. They also cannot be ranked directly against pooled auxiliary inversion, whose reported holdouts used a separate campaign. The intended marker-based timing path remains distinct.

4.4. Sensitivity of auxiliary phase-inversion timing

The strict per-dwell inversion detector achieved approximately 7.47 μs timing RMS among 171 accepted markers out of 800 dwells at 120 dB loss. Availability fell to 57/800 at 124 dB, 5/800 at 126 dB, 1/800 at 128 dB and zero in the tested 130-180 dB cases. These figures describe precision conditional on acceptance; they do not mean that every dwell was timed to microsecond accuracy. Nor do they establish a physical sensitivity limit for the existing reversal waveform [6].

Pooling weak reversal profiles changed the useful operating range. Table 3 shows the eight-case loss screen. At 180 dB, the auxiliary had approximately 19 dB-Hz C/N_0, equivalent to roughly -25 dB per complex 24 kHz sample. No individual dwell qualified there, but all eight accumulated calendar estimates qualified at 36 s. The gain comes from retaining subthreshold evidence, not from extracting a newly precise independent marker on every hop.

Table 3. Pooled-inversion screen, eight matched schedules per loss. All 40 frozen timing intervals covered. These are conditional likelihood intervals with shared clock/model allowances, not universal coverage guarantees.

Loss (dB)	Timing RMS / worst (μs)	Median half-width (ms)	First qualification, median (s)	Whole-phase RMS, nominal → pooled (°)
180	276 / 540	±2.169	8.8	8.592 → 6.054
175	179 / 289	±1.265	6.4	8.076 → 4.325
170	100 / 164	±0.748	6.4	7.898 → 3.742
165	55 / 87	±0.550	6.4	7.833 → 2.281
160	47 / 87	±0.356	6.4	7.813 → 1.492

After screening, the unchanged estimator was evaluated on 32 fresh reserved 180 dB cases. All 32 estimates qualified and their intervals covered: 0.441 ms timing RMS, 0.965 ms worst error and a median conditional half-width of ±2.050 ms, ranging from ±1.919 to ±3.049 ms. Point errors were within 125 μs in 8/32 cases and within 1 ms in 32/32. None of the reported half-widths was within 1 ms. The 125 μs grid and 250 μs bins alone contribute a 187.5 μs discretization

allowance before likelihood and clock uncertainty. This implementation trades unnecessary microsecond precision for sensitivity.

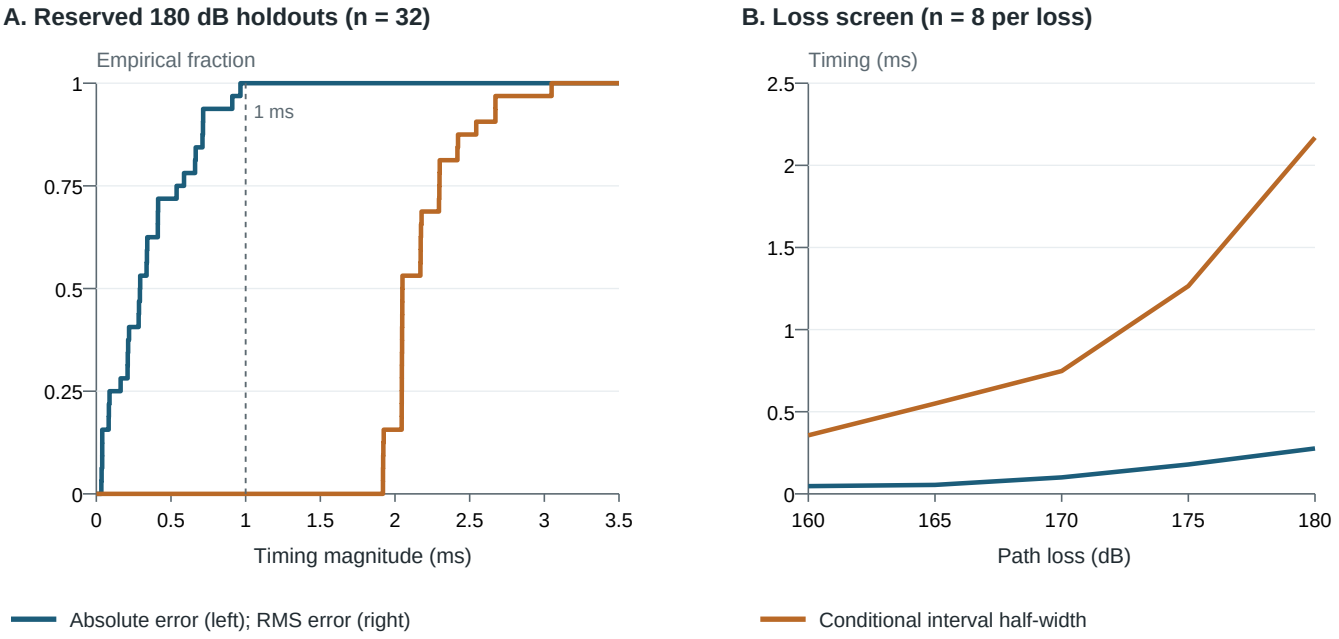


Figure 6. Timing precision and reported uncertainty are different quantities. Left: empirical distributions of absolute frozen timing error and conditional half-width over the 32 reserved 180 dB cases. The 1 ms line is a reference scale, not an acceptance threshold. Right: eight-case loss-screen RMS and median conditional half-width. Connecting lines guide the eye; they do not estimate a physical detection threshold. Data are extracted from the recorded campaign JSON [7].

Initial calendar qualification occurred at 6.4-19.6 s, with a median of 9.4 s. All 2,375 qualified rolling updates covered, and their worst admitted error was 1.964 ms. Those updates share history and are not 2,375 independent trials. A 2 s output buffer cannot repair samples released before this weak-signal acquisition completes. Their initial best-effort correction remains part of the complete-output score.

4.5. Complete output and hop-boundary accuracy

The nominal-calendar and pooled-inversion paths emitted exactly the same 30,721,545 resampled output samples across the 32 held-out captures. Phase and clock correction were applied to all clean output in both paths. Wrong-hop assignments fell from 166,929 to 76,553, a 54.1% reduction. Complete-output clean-twin phase RMS fell by 22.4%, from 8.843° to 6.859°. The unchanged interior accuracy and output count locate the benefit at boundary association rather than selective omission [6,7].

Table 4. Complete-output comparison, 32 reserved HF captures at 180 dB. Uncertain samples remain in the primary metrics; zero-clean-energy samples with undefined phase are counted separately.

Metric	Nominal calendar	Pooled inversion
Emitted samples; phase and clock applied	30,721,545	30,721,545
Conservative well-supported label	6,069,369	6,069,369
Wrong-hop sample assignments	166,929	76,553
Complete-output phase RMS	8.843°	6.859°
Beginning / interior / end RMS	20.074° / 4.5255° / 4.805°	14.045° / 4.5256° / 4.575°
Boundary-region RMS	14.593°	10.443°
Worst complete-output phase error	Approximately 180°	Approximately 180°
Timestamp / reference errors	0 / 0	0 / 0

The result remains far from uniform correction near a hop. Within 0.5 ms of the physical transition, phase RMS was 62.396° after pooling; at 0.5-2 ms it was 38.734°. Figure 7 shows these errors rather than hiding them in an aggregate. The median of each case's boundary p95/p99 changed from 9.0°/78.125° to 7.875°/20.125°; these are medians of per-case percentiles, not pooled percentiles. Phase was undefined at 3,451 zero-clean-energy or pretransmission samples, which were not scored as zero error.

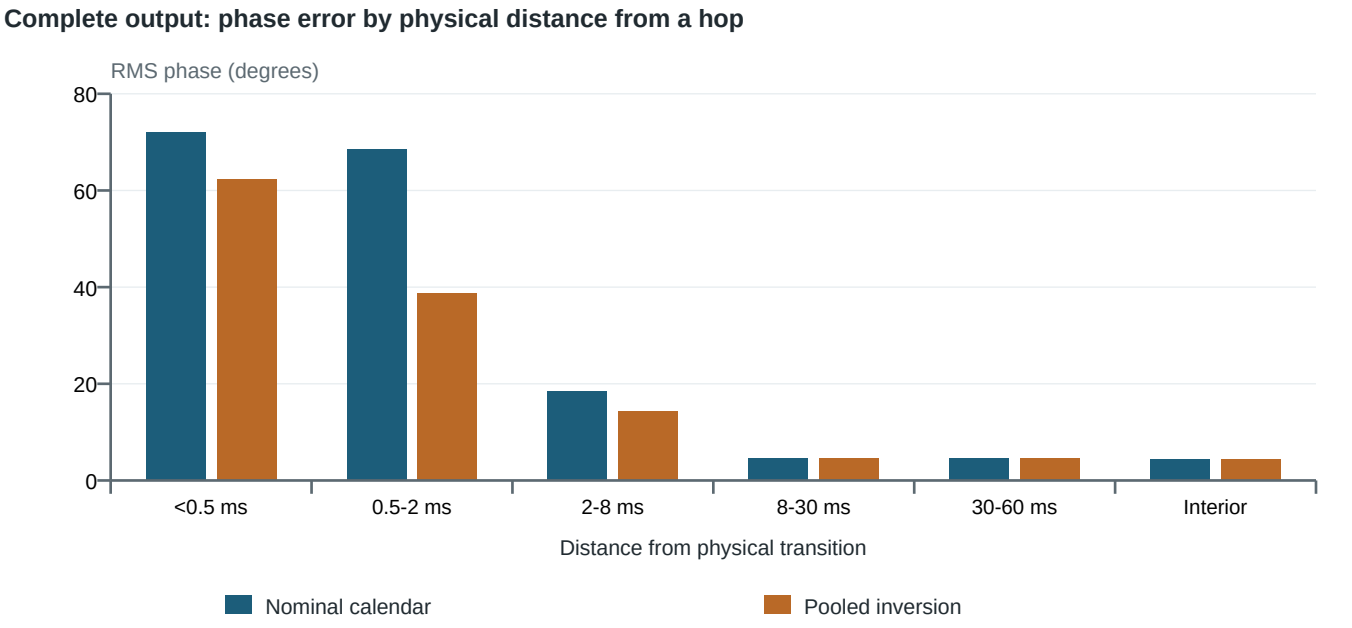


Figure 7. Complete-output clean-twin phase RMS versus distance from the physical hop, including uncertain samples. The unchanged interior and lower boundary error demonstrate a timing benefit without sacrificing delivery. The closest samples remain poorly corrected. Distances and physical transition dates are evaluator-only truth; they do not select receiver corrections [7].

Fresh local pilot phase RMS and worst error remained 2.625° and 12.713° over 3,200 dwells. Maximum evaluated relative-clock error was 19.45 µs, with no observed timing-cell slips in these clean cases. The unchanged frozen frequency result was available in 31/32 cases, and all 31 intervals covered: 0.01357 Hz RMS, 0.02887 Hz worst error

and ±0.11695 Hz median conditional half-width. One case remained unavailable. Pooled timing supplies no independent reason to narrow that frequency interval.

At held-out capture dates 38-40 s, the tone clean-twin phase RMS improved from 5.843° to 3.983°. Boundary fixed-lag per-dwell residual-frequency RMS improved from 0.20287 to 0.07903 Hz; interior residual-frequency RMS remained 0.01320 Hz. The noisy composite sample-phase RMS remained approximately 90°. A separate single-key APSK check at 180 dB improved clean-twin EVM from 5.796% to 5.144%, while noisy EVM changed from 55.321% to 55.516%. At 160 dB, clean/noisy EVM changed from 3.376%/6.276% to 1.855%/5.809%. All 584 evaluable symbols used one fixed reference; the respective five and four interpolation-endpoint exclusions were evaluation limitations, not receiver omissions. These observations do not establish a decoded-data gain.

4.6. Conditional recovery after steps and gaps

A bounded frequency-change detector supplied a provisional acquisition coordinate and a change bracket, not a new observation or correction authority. An actual recovery variant retained raw observations and the original phase-difference reference, then fit one declared additive-common-Hz step while excluding potentially mixed transition apertures. For a 1.2 Hz step after 36 s, 29/32 trials recovered 116/160 later windows, versus 0/160 without recovery. The worst admitted clean-window frequency error was 0.09071 Hz and the worst sample phase was 12.95°, with no admitted slips in those cases. A separately predeclared RF-scaled step model recovered the same number of windows; the detector did not infer the physical step type by selecting whichever model fit best [6].

Extending the detector-only prior from two to three seconds, under a declared 0.002 Hz/s² curvature bound, recovered 124/160 windows and 31/32 epochs. An independent gap option retained the old branch across a 9.6 s interruption and required at least three fresh observations spanning 0.8 s before resuming authority. It restored 126/160 later windows versus 0/160, with worst clean-window frequency error 0.082778 Hz and worst timing error 11.302 µs. The gap crossed the 36 s cutoff, so those cases had no frozen 36 s result.

These gains depended on the model. The gap option assumed bounded audio-clock evolution; eight whole-cycle-jump tests admitted slips in every case, with timing error reaching 545.66 µs and phase error 160.37°. A simultaneous gap and step recovered no windows in eight cases. The options were not established as a qualified combined gap-plus-step solution, nor were all of them requalified in the latest complete-stream path. Their record is useful conditional recovery, not general resilience to arbitrary discontinuities.

4.7. Physical cable evidence and simulation plausibility

An archived non-FHSS analog cable experiment used a 40 s generated stimulus with a +4 Hz common offset and +100 ppm source-clock error. Helper-only conditioning reduced the measured raw tone offset from +4.16018 Hz to +0.000383 Hz and circular phase RMS from 101.48° to 1.098°. The conditional slope-fit standard error was 0.000999 Hz over 30 fixed blocks. This is a physical audio-path measurement with an earlier standalone chain, not an RF loss, FHSS or current complete-stream qualification [10].

A phenomenological simulation of that recording was fitted from pre-probe Helper data and silent preroll, leaving the diagnostic probe out of model fitting. It included gain 0.84150, an 11.34796 Hz first-order high-pass response, 35.0795 ms extra delay beyond the configured preroll, DC offset, white and first-order correlated noise, and ideal signed-16-bit quantization. The correlated-noise time constant was 71.4 ms. With the measured preset, a synthetic capture produced +0.000398 Hz residual tone frequency and 1.0999° phase RMS, closely matching the recorded example without receiver access to the model parameters.

This agreement is evidence that a compact model can reproduce the selected diagnostics of one recording. It is not a unique identification of the physical circuit, an exact replay of its noise, or validation at every frequency and time. A simulated opposite-clock-direction control also exercised negative rate error; it was not a second physical capture. The archived waterfall in Figure 3 uses the original samples and capture dates rather than a visually reconstructed approximation.

5. Comparative tradeoffs and operating limits

5.1. Alternative implementations and design lessons

The successful techniques in Section 4 were developed alongside alternatives that offered mixed or unfavorable tradeoffs. Table 5 records these comparisons to explain implementation choices and guide further work. A local gain may be useful yet insufficient to justify higher cost or poorer complete-stream accuracy. Each result concerns the tested implementation: a Gaussian-summary equivalence test does not settle waveform-level joint estimation, and independently gating short apertures does not test the value of their pooled evidence. Successful pooled inversion timing illustrates that distinction [5,6].

Table 5. Alternative implementations and measured tradeoffs. These comparisons explain retained design choices; offline diagnostics are distinguished from actual-chain tests.

Implementation and scope	Observed outcome	Interpretation
Whole-Hann local frequency refinement; initial eight-case screen	0.02413 Hz point-error RMS and a maximum ±0.29171 Hz guarded half-width; no improvement over the tested baseline.	A bounded whole-aperture refinement was not automatically better than the existing partitioned slope.
Flatter Tukey taper with witness; offline, truth-aided predictions	Training phase RMS 3.17° → 2.83°; held-out 48 ms settling 3.47° → 3.82°. Witness missed 68/127 settling cases and falsely flagged 6/127 clean cases.	Clean efficiency gain did not survive the chosen disturbance test. No taper replacement adopted.
Three short independently gated apertures; feasibility test	Only 3/32 clean dwells acquired, versus 32/32 for the baseline.	Thresholding each short segment discarded useful aggregate evidence; pooled segments remain a different approach.
Soft-rate-v5; eight matched actual-chain clean cases	Phase RMS 2.895° → 2.679°, but clean-tone frequency RMS 0.0180 → 0.0771 Hz; full windows 34 → 16; CPU 0.179 → 0.283 s through 36 s.	A local phase improvement was outweighed by frequency, coverage and cost degradation.
Piecewise-v6, 15 ms bins; four tuning cases	No detections of the 15 ms pulse; 1,840 halo-affected samples retained; worst admitted error 51.90°.	The bounded change model did not reliably identify short physical disturbances at this SNR.
Evolve-v6; one tuning case	Tone-frequency RMS 0.0191 Hz versus 0.0185 Hz; similar pulse/retune errors; no later support after the tested step.	No demonstrated gain from that prior/fresh-likelihood combination; no held-out campaign.
Tighter dynamics interval; evaluator-only curvature test	Errors 0.081335 and 0.096834 Hz exceeded candidate radii 0.078592 and 0.084336 Hz; existing intervals covered.	Smaller modeled intervals were not defensible for the withheld moderate curvature.
Five-boundary ramp feasibility; idealized nuisance conditions	Did not resolve the tested timing-cycle ambiguity at the weak-link operating point.	Failure of that ramp experiment, not proof that all independent timing evidence is ineffective.

The idealized pilot oracle suggested some local headroom: median aided phase RMS of 3.281° compared with 2.693° at known frequency and 2.201° for ideal dual phase with true differential phase. It subtracts exact nuisance waveforms and the per-sample path, knows the true frequencies and differential phase, and initializes frequency search close to truth. It does not supply receiver corrections. Those unavailable advantages make it useful for diagnosis, but neither a practical receiver nor a proven information bound.

Additional local candidates illustrate why the full chain matters. An offline waveform-level joint fit changed frequency RMS from 0.08474 to 0.09281 Hz and phase RMS from 2.703° to 2.714° over 664 common observations; a 0.4 Hz prediction-relation mismatch increased its frequency RMS to 0.16722 Hz while its pre-dwell gate rejected none of 666 attempts. A later split estimator improved second-half phase RMS from 3.995° to 3.653° but worsened first-half frequency RMS from 0.09524 to 0.22831 Hz. Neither updated a validated complete correction chain. These failures do not settle the value of a different uncertainty-aware joint waveform fit.

Pooling short observations also was tested directly, rather than only imposing independent short-segment thresholds. One two-pilot GLRT using 10 ms bins and previous Helper history flagged 1/32 clean cases and only 3/32 short-pulse cases, retaining 10,800 physically disturbed input samples with a worst phase error of 57.06°. Finer 5 ms bins detected 5/32 and retained 10,080 such samples, with 54.72° worst error. The provisional selected spans lacked complete interpolation-halo and post-selection calibration. This specific detector was not promoted, even though pooling weak evidence later succeeded for the known auxiliary reversal.

A separate experimental transmitter timing signature redistributed energy at essentially unchanged total power. Under stable differential phase, a truth-aided two-hypothesis scorer selected branches correctly in eight tested cases; a large differential-phase change produced eight false selections and eight misses. Its decisions were not used by the actual trackers, which still slipped in all eight jump cases. In no-jump actual-chain comparisons, clean APSK EVM worsened from 4.992% to 5.326% on common symbols. The experiment demonstrated conditional information in a changed waveform, not an implemented practical branch recovery.

5.2. Estimator error is not a physical disturbance

A common phase pulse physically rotates the probe as well as the pilots. Estimating the underlying settled oscillator perfectly would not correct that pulse unless the receiver also locates and models it. Conversely, contamination added only to a pilot can bias the inferred correction even though the probe itself remains undisturbed. The transient fixtures distinguish these mechanisms. In the piecewise-v6 48 ms pulse cases, eight events were detected but 2,224 affected samples remained, with a worst admitted phase error of 86.50°. Data-dependent selection, uncertain change dates and interpolation halos remain material.

The pooled calendar experiment reduced wrong-hop counts for previously unseen 23 ms phase-pulse, 51 ms gain-change and 61 ms settling combinations. It did not claim to remove their physical distortions. Thirteen matched impairment pairs produced 896 covered admitted rolling timing intervals, with shared history. At 160 dB, the three additional transient pairs yielded 300 covered individual and 220 covered rolling intervals, but worst individual timing error reached 1.288 ms. Strong clean precision is not a guarantee of equal precision in disturbed dwells.

Twenty-four full-duration controls combined four separately reserved keys, two losses, and straight-auxiliary, absent-auxiliary and noise-only captures. They produced zero individual qualifications, zero pooled qualifications and zero selected boundaries. This is useful finite false-qualification evidence; it does not calibrate a universal false-alarm probability. Estimated noise, a conditioned clock and searched hypotheses make nominal tail settings conditional.

5.3. Marker timing and physical RF retuning can differ

The auxiliary reversal locates the transmitted calendar. A receiver retune can occur independently. With a physical RF retune 2 ms early, the pooled calendar estimate was accurate, yet wrong-hop assignments worsened from 1,224 to 4,976 and complete-output phase RMS from 5.40° to 8.31°. A 2 ms late retune improved from 9,941 to 5,426 wrong assignments but retained its independent displacement. Calendar timing must therefore be combined with known retune latency, an independent RF-transition measurement, or explicit residual uncertainty. An accurate marker does not prove the physical boundary date.

A 9.8 ms arrival outside the ±8 ms search prior selected no boundaries: the edge optimum was not reported as qualified. Curvature and whole-cycle gap tests lost pooled-clock qualification; curvature, gap-cycle and frequency-step cases retained severe held-out phase errors of roughly 116°, 119° and 103°. Arbitrary differential channel phase still can

imitate clock evolution. Delivery continues with available correction and explicit limitations; these cases cannot be called accurately synchronized.

5.4. Correct application of estimates is part of receiver accuracy

The complete-stream work exposed a useful architectural lesson. Two earlier v9 clean cases had interior phase RMS of 42.2° and 34.1° despite Helper clock errors of only about 11 and 8 µs. A missing-phase qualification gate discarded an otherwise valid joint clock point and substituted a broad fallback, adding approximately 417 and 353 µs of mapping error. Repairing that dependency reduced the affected cases to approximately 2.3-2.5°. These were targeted regression repairs, not new independent holdouts [6].

A later provenance repair ensured that a timing profile reports the last Helper samples it actually observed, rather than the end of a larger capture block. An overstated evidence frontier could make the conditioner correctly refuse a correction that was in fact already available. Such failures are distinct from weak-signal estimation limits. Both justify tests of actual sample operations, timestamp semantics and qualification dependencies, not only numerical estimator summaries.

6. Delay, computation and memory

6.1. What additional delay achieved

On one matched HF key, requested delays of 0.4, 1, 2 and 3 s yielded respectively 3,226, 3,117, 3,029 and 2,836 wrong-hop samples. Complete clean-phase RMS was 7.059°, 7.280°, 6.531° and 6.481°. More latency allowed later evidence to improve some still-buffered samples, but did not monotonically improve every metric. This is a single-case comparison, not a statistical optimum for buffer length.

Actual capture-to-release latency was approximately 0.4-1.2, 1.2-1.6, 2.0-2.8 and 3.2-3.6 s because observation and release operate on complete dwells and boundary look-ahead. The frozen 36 s timing estimate was identical at all delays. Original capture-time pattern lookup and reference preservation showed zero errors in the recorded standalone checks. In a missing-ADC case, a 9,600-sample loss remained explicit and all other captured samples were released, including 78 samples with no available phase correction. Neither delay nor interpolation invents lost input.

6.2. Process CPU with exact signal scope

Table 6 uses three serialized repetitions of each method after competing build and simulation jobs finished. CPU includes all receiver threads. Candidate tracking and history, including inversion observation, consumed 0.336-0.374 CPU s for exactly 36 s of capture. This meets the preferred 0.5 s tracking scale in these measurements. It does not meet that scale when complete-stream conditioning and instrumentation are included: at the 36 s frontier, 34 s had been conditioned, and total receiver CPU was 0.592-0.654 s. Across 40 s plus drain, the median was 0.703 s, compared with 0.507 s for the nominal calendar.

Table 6. Serialized CPU measurements. Rows labeled included are already part of the relevant total; overlapping implementation counters must not be added as disjoint costs. CPU seconds are not wall-clock latency.

Component	CPU s	Exact scope
Tracking/history, including inversion	0.336-0.374	Exactly 36 s capture; excludes full-stream conditioning.
Total receiver at 36 s frontier	0.616 median	36 s captured; 34 s delayed input conditioned.
Total receiver through drain	0.703 median	All 40 s; nominal-calendar comparator 0.507 s.
Pilot startup / tracking / fallback	0.0222 / 0.2039 / 0	Guarded pilot observations over 40 s; included.
Inversion observation and pooling	0.1542	100 windows; 7.996083 s retained aperture over 40 s; included.
Full-stream buffer/conditioner/consumer	0.3158	All 40 s plus drain; includes provider and instrumentation.
Profile release queries / buffer push	0.0249 / 0.00142	Subcomponents of the preceding conditioning cost.
Simulation generation	0.3752	40 s capture; excluded from receiver CPU.
Clean-twin conditioning / metrology	0.2820 / 0.4745	Complete evaluated capture and held-out probes; excluded.

The inversion detector uses fixed 267,408-byte storage with at most 96 profiles, 129 timing candidates, three frequency candidates and 384 correlation bins; optimized method stack frames were checked below 16 KiB. Its individual-result fallback ring has 16 records. The complete conditioner allocates 1,847,632 bytes and bounds storage at 115,200 samples in 12 blocks, equivalent to 4.8 s at 24 ksample/s. These are component allocations, not a measurement of total process or evaluator memory.

A measured processing optimization was a bounded 86,400-byte window cache for joint-dwell calculations. All non-CPU fields matched in a 32-seed comparison; mean through-36 receiver CPU changed from 0.34474 to 0.32639 s. Concurrent execution and cache/order effects prevent treating that ratio as a guaranteed speedup. Those earlier 40 s measurements included only partial probe conditioning and are not substitutes for Table 6. No filtering or decimation result in this record establishes additional information or a complete-stream performance gain.

An earlier algebra-preserving covariance optimization retained point estimates, dates and gates across 704 impaired captures, with a maximum diagonal-normalized covariance difference of 1.51×10^{-14}. Five interleaved isolated component benchmarks, each processing 100 settled 0.36 s apertures, reduced median CPU from 0.146559 to 0.133297 s. Startup, conditioning, simulation and metrology were excluded. This result supports an implementation optimization; it is not a whole-receiver speedup claim. Figure 8 places the larger aided-acquisition gain and the cost of the later timing feature in their separate matched contexts.

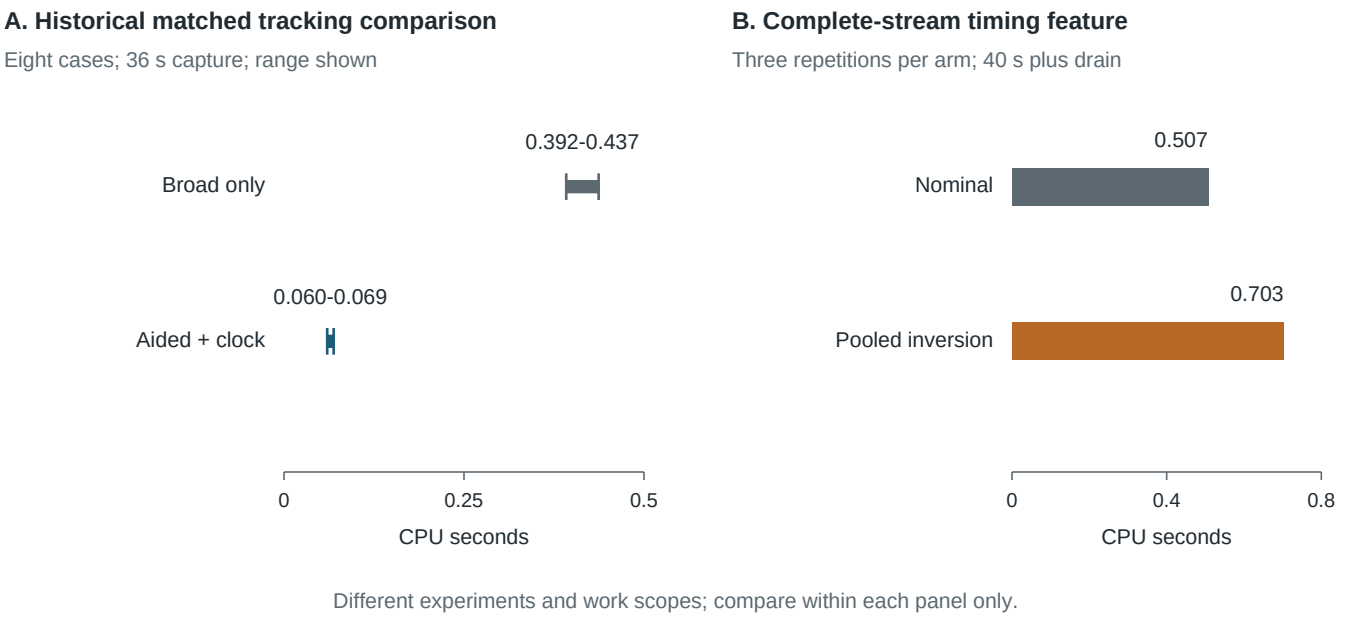


Figure 8. Computational benefit depends on the work being compared. A: recorded range across eight matched historical HF cases for broad-only acquisition versus aided tracking plus relative clock, through exactly 36 s [4]. B: medians of three serialized runs per arm of the later complete 40 s receiver plus drain, nominal calendar versus pooled inversion [6]. The panels have different implementations and processing scopes and must not be read as one chronological speed ranking. CPU is summed across receiver threads; simulation and metrology are excluded.

6.3. Bounded-work interpretation

The timing profile has fixed limits on retained dwells, timing hypotheses and residual-frequency checks. At a structural level, correlation formation scales with the retained aperture samples; evaluation of the bounded profiles scales with the number of bins and tested hypotheses. Complete-stream mixing and two-point interpolation scale with the number of captured/output samples, while buffer storage scales with the maximum retained samples and block records. History length and candidate caps prevent unbounded growth during long reception. These are structural work estimates, not calibrated CPU predictions for a different processor.

Consequently, the measured 0.336-0.374 CPU s over 36 s corresponds to approximately 0.93-1.04% of one CPU core's elapsed-time capacity for tracking. The measured 0.703 CPU s over a 40 s capture is approximately 1.76% of one core-equivalent for the instrumented complete path. Parallel execution may change wall time but does not remove process CPU. SIMD, filtering or decimation should be credited only after matched output checks and direct measurements; fewer samples alone do not establish a more informative or correctly calibrated estimator.

7. Discussion

7.1. What worked best under the principal HF assumptions

The most useful demonstrated combination is frequency-aided guarded pilot acquisition, covariance-aware joint clock/frequency history, dual-pilot fresh phase, selectable actual joint-dwell estimation, and complete-dwell timestamp-preserving conditioning. Pooled auxiliary-inversion timing is a demonstrated recent improvement to weak-link calendar assignment. It uses the existing waveform and tolerates independent random RF phase. Its full-stream gain is accompanied by unchanged output count and essentially unchanged interior accuracy, making the benefit easier to interpret than a support-only improvement.

Frequency assistance achieved useful precision at the priority 40 W / 180 dB condition. The latest available 36 s estimates had 0.01357 Hz RMS point error and a median conditional half-width of 0.11695 Hz. This is a successful

practical outcome at the approximate tenth-hertz scale sought during development. The slightly larger interval retains model allowances; its width is not itself an estimator error or a failed specification. At lower losses, the earlier eight-case joint-dwell screen produced maximum half-widths of 0.0779, 0.0443, 0.0253 and 0.0146 Hz at 175, 170, 165 and 160 dB, respectively. These results establish useful narrowing under the tested assumptions. Curvature and differential-channel counterexamples identify the conditions in which those assumptions need further treatment; they do not negate the demonstrated clean-case performance.

Pooled timing provides useful millisecond-scale calendar information at 180 dB, with a median reported half-width of approximately 2.05 ms. That precision can inform the design of ramp margins and correction boundaries without requiring microsecond timing. At 170 dB the screen median interval was ±0.748 ms; at 160 dB it was ±0.356 ms. The much finer 25 µs and 125 µs examples were possible design scales, not necessary conditions for usefulness. Selection of a downstream margin should use the measured timing uncertainty and retune behavior; actual tolerance remains an integration question.

7.2. Microwave and extreme-loss results are separate evidence

The latest pooled-inversion check included one matched case for each existing 5.8 GHz FCC100 and EU20 wide-hop geometry. Both qualified at 36 s with approximately -1.213 ms timing error inside ±2.117 ms conditional intervals. Whole-stream phase RMS improved from 9.210° to 7.928° and from 9.261° to 7.987°, respectively. These are narrow fixture checks. Their labels do not certify regulatory compliance or establish broad microwave qualification; they also do not reverse the earlier joint-dwell availability losses.

A separate extreme-loss study used a relative audio-clock bound of ±0.0001 ppm, prior removal of deterministic Doppler, a residual-rate allowance of 0.0005 Hz/s and zero actual residual rate [9,11]. At 15 kW and 220 dB loss, eight reduced-mode actual-waveform cases returned covered intervals, but their half-widths were 0.2743-0.3617 Hz despite a maximum point error of only 0.0333 Hz. At 1.5 kW and 220 dB, none of eight met the study's reduced-mode criterion. A four-seed path-loss refinement reached 204.5 dB at 15 kW and the C/N_0-equivalent 194.5 dB at 1.5 kW for that study's ±0.1 Hz screening threshold. This threshold defined the recorded comparison, rather than a general adequacy requirement. These small, condition-specific results are neither independent proof at two sensitivities nor a physical limit. They do not test the current pooled timing chain at 220 dB or establish an Earth-Moon-Earth link.

7.3. Threats to validity and remaining work

The experiments assume quiet enough spectrum that the strongest tone in the oscillator-derived high-pilot range can be identified as the pilot. They include selected leakage, transient and noise controls, but do not characterize every interferer, fading channel or multipath environment. Thermal temperature and noise figure are explicit simulated inputs, not hardware measurements. Empirical coverage from 8 or 32 cases, and especially from correlated rolling updates, must not be promoted to a universal probability guarantee.

The complete conditioner was evaluated chiefly around the 1600 Hz tone and 360 Hz shaped APSK band. Its current linear interpolator is not transparent over the entire 24 ksample/s Nyquist band: a separate deterministic check reported approximately 1.94% interpolation EVM at 1780 Hz and 60% at 10,800 Hz under a 100 ppm resampling case. A wider-band downstream contract requires an appropriate resampler and new tests, while preserving capture-time mapping and discontinuities [3,6].

Some comparisons mix different campaign versions only as historical context. The latest embedded v8 common-support diagnostic was available in just 2/32 cases, totaling 42,078 samples; it is not a comprehensive old-versus-new common-support study. The principal timing comparison is stronger in a different way: both methods emit the same complete stream and all uncertain samples remain scored. The EVM and delay variants have much smaller sample counts than the 32-case timing comparison.

No new receiver experiment was run to prepare this article. It synthesizes the recorded development evidence and computes a spectrogram of an existing recording. Section 4.7 presents the earlier physical cable evidence [10]; the latest pooled timing and complete-stream results remain simulations, not new audio-loopback or over-the-air measurements.

No ordinary Robust integration, successful Robust decoding, regulatory certification or universal sensitivity boundary is claimed. Promising future work includes independently measuring RF transition timing, a practical transient model that improves complete output, defensible explicit dynamics, and a bandwidth-appropriate timestamp-preserving resampler. The recorded alternatives leave scope for further useful development.

8. Recommended receiver design practices

The following recommended practices follow from the successful receiver experiments and their comparative evaluation. They provide a practical starting design under the stated signal, clock and channel assumptions. The recommendations distinguish the broadly useful receiver architecture from additions suited to particular operating conditions. Evidence from older guarded chains is identified separately from validation of complete-stream behavior.

Table 7. Recommended practices, demonstrated benefits and conditions of use.

Practice	Supporting result and rationale	Conditions and limits
Retain one relative time/phase reference.	Two-pilot phase differences suppress common random RF hop phase; relative-clock tracking improved fixed-reference output when aided phase alone did not.	Constant differential phase remains in the initial reference. Changing differential phase and whole-cycle jumps are not generally identifiable.
Use frequency-aided correlations with broad acquisition and fallback.	The earlier matched eight-case study substantially reduced tracking CPU while preserving a recovery route for inconsistent predictions.	Requires valid acquisition coordinates, dated observations and leakage protection; the quiet-spectrum assumption still applies.
Combine pilot information with its covariance.	Joint history and dual phase improved clean HF tracking; actual joint-dwell fitting further improved common-support phase and availability.	Predictions and reused samples are not independent evidence. Microwave availability losses prevent a universal preference for the joint-dwell option.
Buffer and correct complete dwells; carry capture-time metadata.	Settled measurements can correct earlier and later samples of their own hop. Complete delivery and timestamp tests expose errors hidden by support masks.	Do not interpolate across missing input or smooth random RF phase across hops. The current resampler has a limited demonstrated bandwidth.
Pool known weak timing structure when individual markers fail.	Auxiliary profiles provided useful calendar estimates at 180 dB without individual marker qualification; ramp tracking separately improved boundaries when ramps existed.	Use explicit shared clock/model uncertainty. Calendar timing does not independently establish RF-retune timing; acquisition latency remains.
Treat recovery and uncertainty as model-conditional.	Explicit step/gap recovery restored useful output in declared cases, while curvature and cycle-jump controls exposed failures.	Use fresh evidence to authorize recovery, preserve reference, and retain unavailable/uncertain results. Choose uncertainty allowances from the operating model, independently of an aspirational target.
Evaluate complete output and measure complete work.	Common-support checks, clean/noisy twins, held-out keys and full-stream CPU accounting distinguish estimator gains from selection and scope effects.	State symbol/sample exclusions, signal durations and all-thread CPU. Noisy EVM, oracle error and receiver error answer different questions.

For the principal HF condition, the practical starting combination is guarded aided tracking, covariance-aware relative clock/frequency estimation, dual-pilot fresh phase, and the complete timestamp-preserving conditioner. Current-dwell joint estimation and pooled inversion timing are selectable additions with measured benefits and costs. Ramp tracking,

gap extension and step recovery warrant retention as explicitly conditional alternatives rather than silent defaults across every operating model.

Before downstream integration, the bandwidth/resampling contract, capture-clock epochs, independent RF-retune latency and pattern-selection timestamp interface need explicit agreement and tests. Complete baseband delivery should not depend on meeting an exact ±0.1 Hz interval. Delivery, correction availability and correction confidence are different outputs; preserving them separately is one of the clearest practical lessons of the development record.

9. Conclusion

The standalone Helper Carrier development produced effective, computationally inexpensive techniques for frequency assistance, fresh-hop phase tracking, relative-clock estimation and complete baseband delivery. At 40 W / 180 dB, the latest available 36 s frequency estimates achieved 0.01357 Hz RMS error with a median conditional half-width of 0.11695 Hz: useful narrowing at the approximate scale sought during development. Joint clock history and dual-pilot phase improved clean-hop tracking, while actual joint-dwell estimation added a further HF gain. Complete-dwell conditioning applied the estimates across captured dwells and preserved their original timestamps. Pooling auxiliary-reversal evidence achieved 0.441 ms timing RMS, 54.1% fewer wrong-hop assignments and 22.4% lower complete-output clean phase RMS without dropping output.

The recommended design combines aided acquisition, covariance-aware use of both pilots, a permanent relative reference, complete-dwell buffering and explicit capture-time metadata. Known timing structure should be pooled when individual markers are too weak, and recovery should use fresh evidence under a stated evolution model. Tracking required 0.336-0.374 CPU s per 36 s; the instrumented complete path required a median 0.703 CPU s over 40 s plus drain. The remaining engineering work concerns samples immediately around physical retunes, transient disturbances, resampler bandwidth and unresolved differential-phase or whole-cycle ambiguities. These are specific limits on applying otherwise useful techniques. The results provide a demonstrated standalone foundation for integration; downstream decoding remains to be tested.

Appendix A. Evidence and reproduction

The result figures use extracted recorded data; format and ideal-spectrum illustrations are explicitly schematic. The waterfall is newly computed from the archived cable capture, whose decompressed SHA-256 is cd15730de6c0b67b2ce69f2e08e9d6c33fdef7724f5ec2a7ed074f819a56a405. No receiver thresholds or source were changed. The accompanying source bundle contains the editable manuscript, scientific figures, renderer, extracted timing and spectrogram data, and evidence manifests with hashes. The original source checkout inspected at authoring was 147dbc94f1709a7c563814b3d384326bb0e2dfbc; revision-specific inspected inputs are identified separately. That checkout identifier does not substitute for the campaign binary hashes.

The loss/holdout campaign used binary SHA-256 df88e4cfa9c93c7338ad15feed3fca6a4f561daf796c68990425d77c6c8ece98. Later accounting-only changes used 049d8b01b1c41df9260c8d58c5b15accd46af73d4710c1839770ae8e90b16e67. The recorded same-case nominal/profile comparison preserved every pre-existing non-CPU output field after that accounting change. Archived source and binaries are needed for exact historical reproduction; current commands reproduce the available implementation, not automatically every earlier revision.

The recorded validation reports 17 focused Helper checks passing, plus focused ASan/UBSan/LeakSanitizer checks for the inversion profile and complete conditioner. The initial sandboxed leak-sanitizer invocation could not operate under the process tracer; the same synthetic tests subsequently passed with leak detection retained outside that restriction. These are reported development results, not tests rerun while writing this paper. The ordinary Robust regression campaign was outside their scope [6].

```
# Focused recorded build/check entry points, from repository root
CCACHE_DIR=/tmp/pumpmodem-helper-ccache ./build.sh configure
CCACHE_DIR=/tmp/pumpmodem-helper-ccache cmake --build build/dev \
  --parallel 2 --target test_helper_inversion_profile \
  test_helper_complete_study test_helper_transient_fixture
ctest --test-dir build/dev --output-on-failure --parallel 1 \
  -R '^(helper_inversion_profile|helper_complete_profile_regression|helper_transient_fixture)$'

# Matched complete-chain comparison; 40 W Helper + 40 W probe
build/dev/test_helper_complete_study full 180 1930100001 \
  clean tone 2 hf nominal
build/dev/test_helper_complete_study full 180 1930100001 \
  clean tone 2 hf pooled-inversion
```

The loss screen uses seeds 1930000001-1930000008 at each of 180, 175, 170, 165 and 160 dB. The 32 held-out cases use 1930100001-1930100032 at 180 dB; they are now regression inputs. Noise seeds are offset by 20,000,000. Replace tone with apsk for the probe, delay 2 with 0.4, 1 or 3, and hf with microwave or microwave-eu for the fixture variants. Recorded impairment and control modes include straight-auxiliary, absent-auxiliary, noise, and the reserved transient-inversion-held-phase23, gain51 and settle61 variants; consult the full command definitions in [6] before invoking them.

References

[1] D. C. Rife and R. R. Boorstyn, “Single tone parameter estimation from discrete-time observations,” *IEEE Transactions on Information Theory*, vol. 20, no. 5, pp. 591-598, 1974. doi:10.1109/TIT.1974.1055282.

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[9] pumpModem project, *Helper EME measurement*, docs/helper-eme-measurement.md, reduced-mode 220 dB and path-loss refinement experiments, inspected 31 October 2026.

[10] pumpModem project, *Helper live probe*, docs/helper-live-probe.md, related standalone probe methodology and earlier live-test context, inspected 31 October 2026.

[11] pumpModem project, *Helper measurement*, docs/helper-measurement.md, oscillator and link-model background, inspected 31 October 2026.

[12] pumpModem project, *Helper Carrier*, docs/helper-carrier.md, waveform, lead-in and operating assumptions, inspected 31 October 2026.